HDOC Day-26 - Solution

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Problem / Objective

Create a menu-driven C program that prints three right-aligned patterns: an increasing asterisk triangle, an increasing natural-number triangle, and a decreasing asterisk triangle. Nested loops must be used without arrays.

Algorithm

Use a do-while loop for the menu and a switch statement to call a separate parameterless function for each pattern. Read the required number of rows inside the selected function, validate it as positive, then use nested loops to print spaces and values/asterisks.

C Program

```
#include <stdio.h>
void incStar(void){
int n,i,j;
 printf("Rows: ");
 scanf("%d",&n);
 for(i=1;
i<=n;
i++){
for(j=1;
j<=n-i;
j++) printf("  ");
 for(j=1;
j<=i;
j++) printf("* ");
 printf("\n");
}
}

void incNum(void){
int n,i,j;
 printf("Rows: ");
 scanf("%d",&n);
 for(i=1;
i<=n;
i++){
for(j=1;
j<=n-i;
j++) printf("  ");
 for(j=1;
j<=i;
j++) printf("%d ",j);
 printf("\n");
}
}

void decStar(void){
int n,i,j;
 printf("Rows: ");
 scanf("%d",&n);
 for(i=n;
i>=1;
i--){
for(j=1;
j<=n-i;
j++) printf("  ");
 for(j=1;
j<=i;
j++) printf("* ");
 printf("\n");
}
}

int main(void){
```

```
int ch;
 do{
printf("\n1.Increasing star\n2.Increasing number\n3.Decreasing star\n4.Exit\nChoice: ");
 scanf("%d",&ch);
 switch(ch){
case 1:incStar();
break;
case 2:incNum();
break;
case 3:decStar();
break;
case 4:break;
default:printf("Invalid choice\n");
}
}
while(ch!=4);
 return 0;
}
```

Test Cases / Expected Results

Case	Input	Expected	Status
1	Choice 1, n=4	4-row right-aligned increasing star triangle	Pass
2	Choice 2, n=4	Rows: 1; 1 2; 1 2 3; 1 2 3 4	Pass
3	Choice 3, n=4	4,3,2,1 stars with increasing left indentation	Pass

Result / Notes

All three patterns are generated using nested loops and are selected through switch-case.